

Prem Mallappa

Systems Software Architect • Embedded Systems • Virtualization

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Seasoned Software Architect with extensive expertise in designing and developing high-performance embedded systems, system software, Linux kernel drivers, and virtualization technologies. Proficient in a wide range of programming and scripting languages, with a proven track record of delivering robust, scalable solutions in complex technical environments. A highly analytical and collaborative team player, recognized for strong problem-solving abilities, logical thinking, and a passion for mastering emerging technologies. Committed to driving innovation and excellence through clean architecture, optimized code, and best-in-class engineering practices.

Skills

:PROPERTIES: :CV_{ENV}: cvskills :END:

Leadership 22+ years of multidisciplinary team and project leadership experience; software architecture and technical direction across systems and platform teams.

Communications Clear written and spoken communication for architecture reviews, design discussions, mentoring, and cross-functional execution.

Technology Broad and deep IT expertise across embedded systems, system software, cloud/security, virtualization, and software services development.

Computer Security Cryptography, virtualization and cloud computing security architecture, secure software development, risk management, compliance, intrusion detection, and prevention.

Programming Languages C, C++, Assembly (x86, ARM, MIPS), Rust, Python, Go, BASH

Operating Systems Linux, FreeBSD, Symbian

RTOS VxWorks, QNX, seL4, Zephyr, ThreadX

Architecture x86, ARM v6/v7/v8, MIPS64, PowerPC, RISC-V

Development Tools GCC, Clang, Git, CMake, Make, GDB, Perf, VTune, ARM Development Tools, Keil, JTAG, Docker, QEMU

Systems & Virtualization Linux kernel development, device drivers, hypervisors, IOMMU/SMMU, pre-silicon validation, boot flows, and low-level firmware bring-up.

Performance & Acceleration SIMD/vectorization, performance optimization, math and crypto libraries, SmartNIC/DPU offload, NVMe-oF, RDMA, and hardware-assisted data paths.

Specialized Skills Embedded systems, system software, cryptography, filesystems, performance engineering, packet/data-path acceleration, and platform enablement.

Education

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Masters (Computer Science)

BITS Pilani

Pilani, Rajasthan, India

2016

Bachelors (Computer Science)

Adichunchangiri Institute of Technology (AIT)

Chikkmagalur, Karnataka, India

2002

Affiliated: Visvesvaraya Technological University (VTU)

Open Source

:PROPERTIES: :CV_{ENV}: cventries :END:

QEMU [\(Link\)](#)

2014-2016

- SMMUv3 (IOMMU) emulation support for ARMv8
- Designed and implemented SMMUv3 model merged into mainline
- Added support for Stage1, Stage2, and nested virtualization
- Implemented command queue processing and page table walk

Linux Kernel (IOMMU/SMMUv3)

2015-2016

Contributed to upstream SMMUv3 driver for Broadcom Vulcan

- [Fix GERRORN register write](#)
- [Fix STE.VALID check in Vulcan](#)
- [Workaround for STE.VALID](#)

Linux Kernel (Kexec/Kdump) [\(Link\)](#)

2011-2013

- MIPS Kexec/Kdump port for Cavium Octeon platforms
- Developed MIPS64 port of Kexec and Kdump (merged upstream)
- Fixed critical bugs in KEXEC for Cavium Octeon platforms

GLIBC [\(Link\)](#)

2018-2020

- Performance optimizations for AMD processors
- Fixed memcpy behaviour on AMD processors
- Optimized string functions for x86-64 architecture
- Performance improvements for memory operations

AMD LibM [\(Link\)](#)

2018-2025

- Open source math library for AMD processors
- Core contributor to open sourcing AMD Math Library
- Optimized transcendental functions (exp, log, pow, trig)
- Implemented SIMD/FMA optimizations for vector operations

Experience

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AMD Ltd.

2018 – Present

Bengaluru, India

PRINCIPAL ENGINEER

Worked in multiple teams within AMD including **AMD LibM**, **AOCL Cryptography**, **BSP**, and **NVMe** in **Pensando** BU. Every project was different, requiring quick learning and adaptation to new domains. The role also involved **leading a team** of four-to-six engineers in various projects. I was also part of the leadership team where I mentored engineers from across AOCL and Pensando BU.

SR. MEMBER OF TECHNICAL STAFF

Worked on **Model0** which was **pre-silicon validation** of next generation AMD cores, including Zen2 and Zen3. The role involved working with hardware team to bring up bare-minimum processor configuration with just core and memory controller, and run Linux, stressors and benchmarks. This involved working on low-level firmware, bootloaders, Linux kernel and performance analysis tools.

Broadcom Ltd.

2014 – 2016

Bengaluru, India

PRINCIPAL ENGINEER

Architected foundational software stack for Broadcom Vulcan, a multicore-multithreaded ARMv8 64-bit server processor. Engineered complete **SMMUv3** (System Memory Management Unit) driver infrastructure supporting advanced IOMMU virtualization. Developed and contributed an **SMMUv3 emulation model** to QEMU mainline, enabling early platform validation and ecosystem enablement. Implemented comprehensive virtualization features including command queue processing, STE/CD descriptor parsing, multi-level page table walks, and stage-1/stage-2 address translation.

Cavium India Pvt. Ltd.

2011 – 2013
Bengaluru, India

TECH LEAD

Led low-level software development for Cavium Octeon III (MIPS64) multicore network processors, delivering stable platform firmware for networking customers. Resolved a critical KEXEC kernel bug and authored the full **MIPS64 port** of the Kexec/Kdump crash dump infrastructure, ultimately merged into Linux mainline. Engineered a bare-metal core dump generation framework with a host-resident daemon that enabled detailed post-mortem GDB analysis during bring-up. Designed and implemented the **CavHv hypervisor** for MIPS64 architecture using hardware virtualization extensions to validate next-generation features.

ARM Ltd.

2005 – 2009
Bengaluru, India

SR. DEVELOPMENT ENGINEER

Executed comprehensive **OS porting initiatives** for emerging ARM architectures including ARM1176JZFS with TrustZone security extensions and the Cortex-A8 application processor. Developed a production-grade touchscreen driver for Symbian OS along with a Python-based automated validation framework used across multiple programs. Designed a precision interrupt latency measurement infrastructure that quantified TrustZone secure-world context switching overhead for silicon partners. Ported the **L4Ka::Pis-tachio microkernel** to ARM architecture, enabling virtualization research and multiple proof-of-concept implementations.

Sasken Communication Technologies Ltd.

2004 – 2005
Bengaluru, India

SR. SOFTWARE ENGINEER

Architected and maintained the **Extended File System (EFS)** for VxWorks-based UMTS base station infrastructure, balancing throughput with resiliency requirements. Engineered a reset-resilient filesystem with **advanced wear-leveling algorithms** so data integrity held across unexpected power cycles in the field. Designed a flash-optimized flat file tree structure that minimized write amplification and significantly extended device longevity.

Global Edge Software Ltd.

2003 – 2004
Bengaluru, India

SOFTWARE ENGINEER

Designed and implemented a **comprehensive SDIO host controller driver** for Linux on Intel StrongARM embedded platforms used in consumer devices. Engineered a high-performance 4-bit SDIO driver for Marvell 802.11g WiFi chipsets that consistently achieved maximum theoretical throughput. Optimized **DMA transfers** and interrupt handling paths to minimize latency and maximize wire-less data transfer efficiency under heavy load.

Short Stints

VSPL Ltd.

Software Architect

Nov. 2016 – Jan. 2017
Bengaluru, India

Cisco Ltd.

Software Engineer

Oct. 2010 – Apr. 2011
Bengaluru, India

B-Labs, London UK

Sr. Engineer - Contractor

Nov. 2009 – Sep. 2010
Bengaluru, India

Harman International

Engineer

Jul. 2009 – Nov. 2009
Bengaluru, India

Personal Projects

HELM — Heterogeneous Execution and Latency Model

Sep. 2026 – Present

Synopsis	Research-grade hardware simulator targeting the same problem space as gem5 , QEMU , and Simics , but designed from first principles for clarity, correctness, and composability. Runs AArch64 and RISC-V RV64GC binaries in syscall-emulation or full-system mode with selectable timing fidelity, built as a Rust core behind a Python-first control plane.
Details	Architected a 27-crate Rust workspace spanning framework, runtime, hardware, and debug layers. Designed the engine so that timing is the sole generic parameter , monomorphized and inlined into the engine rather than dispatched through a vtable, while ISA and execution mode are enum-dispatched once per Python call , costing nothing per instruction. Implemented three timing models spanning functional (IPC=1), interval estimation over a two-level L1D/L2 cache hierarchy , and cycle-accurate. Built a Device trait and bus tree with dynamic device loading, where platform code owns base addresses and IRQ wiring instead of the device. Exposed a gem5-style Python configuration layer via PyO3 as the first-class interface for configure, start, stop, query, and debug. Enforced determinism by default , admitting no wall-clock reads and no background threads in the hot loop.

Link:
[Project](#)

Languages:
Rust,
Python, C

Tools:
Cargo, PyO3,
mdBook,
Clippy, Git

Tech:
AArch64,
RISC-V
RV64GC,
Syscall
Emulation,
Full-System
Simulation,
Cycle-
Accurate
Timing,
Device
Modeling

zclip — Zellij Yank-Buffer Plugin

Sep. 2026 – Present

Synopsis	Zellij plugin bringing tmux-style paste-buffer semantics to a terminal multiplexer that lacked them, compiled to WebAssembly against the zellij-tile plugin API. Alpha software with the core yank, paste, and copy-mode workflow working end to end.
Details	Implemented a paste-buffer ring with bounded eviction alongside named, pinned buffers that are never evicted, mirroring tmux semantics. Built a copy mode over terminal scrollback supporting vi and emacs keymap presets with per-action key overrides, cursor motions, and char, line, and block selection rendered with reverse-video highlighting. Developed a pipe interface (yank, paste, copy mode, cancel, list) drivable from keybindings or zellij pipe , plus headless background loading so paste targets whichever pane currently holds focus. Added opt-in per-session persistence of the buffer ring surviving plugin reloads and detach/attach, with the privacy tradeoffs documented. Packaged via a Nix flake and backed by CI running fmt, clippy, tests, and the wasm build, with a tag-triggered release workflow.

Link:
[Project](#)

Languages:
Rust,
Python, Nix

Tools:
Cargo, Just,
Nix Flakes,
GitHub
Actions, Git

Tech: We-
bAssembly,
wasm32-
wasip1,
Zellij Plu-
gin API,
Terminal
Scrollback,
KDL Config-
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